



VAULTX CYBER TECH

BEYOND TODAY · BEFORE TOMORROW

SUMMER INTERNSHIP PROGRAM 2026

WEEK

01

AI FOUNDATIONS & ENVIRONMENT

PYTHON ENVIRONMENT / LLM FUNDAMENTALS / TOKENS & CONTEXT / API INTEGRATION / GIT WORKFLOW

6

TASKS

6

DELIVERABLES

BEGINNER

LEVEL

ONSITE / LAB

MODE

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WEEK FOCUS

Objective	Get a working development environment, understand what a language model actually is, and ship your first program that calls one.
Skills Targeted	Python environment LLM fundamentals Tokens & context API integration Git workflow
Tools & Stack	Python 3.11+ VS Code Git / GitHub venv Anthropic or OpenAI SDK python-dotenv

TASK LIST

01 Set up a professional dev environment

- Install Python 3.11+, VS Code with the Python extension, and Git.
- Create a virtual environment and a requirements.txt. Never install packages globally.
- Create the GitHub repo vaultx-ai-internship with a README and a proper .gitignore.
- Make at least five meaningful commits this week — one giant commit at the end scores zero.

02 Build the vocabulary

- Define in your own words: token, context window, temperature, top-p, system prompt, embedding, hallucination, fine-tuning, inference.
- Explain the difference between a base model and an instruction-tuned model.
- Explain in three sentences why an LLM predicts text rather than 'knows' facts.

03 Make your first API call

- Get an API key and store it in .env; load it with python-dotenv.
- Write a script that sends a prompt and prints the response.
- Print the token usage and calculate the cost of that single call.

04 Experiment with generation parameters

- Run the same prompt at temperature 0, 0.7 and 1.0, three times each.
- Record the outputs and note how determinism changes with temperature.

05 Build a reusable API wrapper

- Wrap the API in a module with error handling: retries, timeouts and clear exceptions.
- Everything you build later will import this module.



06 Ship a working mini-tool

- Build a CLI tool that takes text and returns a summary, key points and sentiment.
- It must accept input from a file or a command-line argument.
- Record a demo video showing it running on at least two different inputs.

DELIVERABLES

#	DELIVERABLE	FORMAT
1	GitHub repo with venv, requirements.txt and .gitignore	REPO LINK
2	AI terminology glossary (9 terms, own words)	PDF / MD
3	First API-call script with token & cost output	PY
4	Temperature comparison table with analysis	PDF
5	Reusable API wrapper module with error handling	PY
6	CLI summariser tool + demo video	PY + MP4

SUBMISSION PROTOCOL

Channel	Upload to the VaultX LMS under Internship > AI Track > Week 01.
File Naming	VX_AI_W01_FirstName_LastName.pdf plus one ZIP with all supporting files, exports and screenshots.
Deadline	Sunday, 11:59 PM (PKT) of the assigned week.
Evidence	Every screenshot must show your machine clock and the relevant page/scope.
Walkthrough	Record a short screen video (3–5 min) explaining what you did, what you found, and how you stayed safe.